

Munaza Ashraf

Graduate Research Applicant in World Models and Physical AI

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Education

Sep 2020 **B.Sc. (Hons.) Software Engineering, University of Engineering and Technology (UET)**
Jun 2024 *Taxila, Pakistan*

- CGPA **3.94/4.0**, **Silver Medalist** (second highest aggregate across all four years).
- **Final Year Project: PsycheSync**, an AI mental health platform with mood tracking, journaling, assessments, and a moderated peer community (Flutter).
- Fine tuned **LLaMA 2** on medical guidance for precautions, remedies, medicines, prescriptions, and meditation, and fine tuned **OpenAI GPT-2** to role play as a psychiatrist.

Research Interests

Areas World Models, Physical and Embodied AI, Robot Learning, Multimodal and Generative Models, Computer Vision and 3D Vision, Knowledge Distillation, Efficient Training and Inference

Direction Developing efficient **world models** for **physical AI**: models that learn spatial, temporal, and physical dynamics from multimodal data, transferred into compact models and policies through **knowledge distillation** and efficient training and inference under strict latency, memory, and compute constraints.

Publications & Preprints

2026 **Recovering a Withheld Audio Tokenizer from Its Generator: Reverse Distillation of the MiniMax Music 3 Encoder, In and Out of Distribution**
Under Review

Munaza Ashraf, bghira. Under review, 2026. [Code] [Models] [Dataset]

- Recovered the withheld tokenizer of the open 8B MiniMax Music 3 generator by **reverse distillation**, giving a text only model audio prompting: reference guided generation and continuation.
- The encoder **transfers out of distribution**: recorded music keeps as much reachable conditioning as generated music (**0.92 vs. 0.91**), and degraded view training restores robustness to lossy audio (**0.82 vs. 0.50**).
- Released the encoders, code, and **MM3 OOD**, a new **240** track out of distribution benchmark.

Independent Research & Reproductions

Aug 2026 **WPT: World-to-Policy Transfer via Online World Model Distillation**

Sep 2026 *Code implementation of the published paper. [Paper] [Code]*

Tools: OccWorld, nuScenes, PyTorch

- Implemented and open sourced the paper's method on a frozen **OccWorld** world model: rollouts, trajectory proposal, reward scoring, and **policy distillation** into a compact student.

May 2026 **Head Wise Adaptive Sparse Attention for Video Diffusion, HASTE**

Tools: Wan 2.2, LTX 2.3, Sparse Attention

- Implemented training free video diffusion acceleration with temporal mask reuse and error guided budgeted calibration using per head top p thresholds.

2025 **Diffusion Inference Caching Benchmark**

2026 **Tools:** FBCCache, TeaCache, MagCache, Cache DiT, Fast3D Cache, Nsight Systems

- Benchmarked caching methods across image, video, and 3D generative models, measuring kernel time and peak VRAM against FID, CLIP score, SSIM, and LPIPS.

- May 2026 **Efficient Fine Tuning and Optimizer Studies**
- Jul 2026 **Tools:** TREAD, DreamBooth, LoRA, ConsisLoRA, SOAP, Muon, AdamW, Lion
- Studied **TREAD token routing** with LoRA and context parallelism, measuring convergence, memory, throughput, and quality.
 - Ran controlled **optimizer comparisons** (SOAP, Muon, AdamW, Lion) for diffusion, LLM, and VLM adapters.
 - Designed **segmented gradient checkpointing** for Flux and MageFlow, giving a finer memory versus recompute trade off and lower step latency.

Research and Work Experience

- Nov 2024 **Applied ML Engineer, Runware, London, UK (Remote)**
- Present **Tools:** PyTorch, Diffusers, Transformers, vLLM, Go, CUDA, ROCm, NCCL, FlashAttention, SageAttention, FlashInfer, Docker, Prometheus, Vercel
- Integrated, optimized, and served **1,000+** image, video, audio, 3D, LLM/VLM, and multimodal models in production, with up to **2×** faster inference on selected workloads.
 - Built model specific schedulers and execution recipes with batching, VRAM aware placement, and hot swapping across model families: **image** (FLUX, Stable Diffusion, SAM, Ideogram, Anima), **video** (LTX, Wan, HunyuanVideo, Ovi, VEED Fabric 1.0, MiniMax), **audio** (ACE Step, Dia, Mirelo), and **3D** (TRELLIS, SAM 3D).
 - Designed **distributed and serverless inference** across NVIDIA RTX, Hopper, and Blackwell GPUs (RTX PRO 6000, H100, H200, B300) and AMD ROCm, using context parallelism with NCCL communication overlap.
 - Benchmarked FlashAttention, SageAttention, FlashInfer, and cuDNN SDPA per GPU architecture to route each model to its **fastest attention backend**, combined with diffusion and KV caching and **FP8, FP4, INT8, INT4** quantization.
 - Worked on **teacher student knowledge distillation** for compact, faster model variants.
 - Trained LoRAs once on the base model and deployed them on **base, distilled, and turbo variants** by finding the transformer layers with the highest variance or modulation shift between variants and skipping or bridging them, so the adapter converges well on every variant.
 - Served high volume enterprise workloads for **Higgsfield, Freepik, Wix, PixVerse, Civitai, and the LTX team**, with API and frontend integrations deployed on Vercel.
- Sep 2024 **Founding Engineer, MedTalk, Faz Australia, Australia (Remote)**
- Jul 2025 **Tools:** Whisper, AssemblyAI, MedPaLM, LLaMA, RAG, Pinecone, Vector Search, WebSockets, AWS, RunPod
- Founding engineer for **MedTalk**, a real time clinical AI platform used by **50,000+** clinicians; owned ML, backend, and production ML infrastructure.
 - Curated and prepared clinical training datasets, and fine tuned speech, medical NLP, and language models including **MedPaLM** for structured clinical note generation.
 - Cut post VAD streaming speech latency from **~150ms to ~50ms P50 (3×** faster) on Whisper and then AssemblyAI, with WebSocket streaming and optimized model serving.
 - Built a **RAG based clinical assistant** on Pinecone vector search, grounding LLM answers in retrieved medical context.
 - Built a **retrieval augmented billing code LLM** that maps consultations to medical billing codes, deployed on AWS.
 - Implemented privacy and consent controls for HIPAA, Australian Privacy Principles, and PIPEDA.
- Jun 2024 **Computer Vision Engineer, VisionRD, Islamabad, Pakistan**
- Nov 2024 **Tools:** YOLO, OpenCV, PyTorch, Real Time Video Pipelines
- Built a **wearable AI assembly copilot** for Toyota (head mounted 4K camera, real time error detection, cycle analysis), improving line efficiency by **~35%**.
 - Built automated worker activity and assembly validation systems for Thal Engineering.
 - Developed low latency visual defect detection for KIA Motors to automate quality inspection.
- 2022 **STEM Trainer, Junior Coderz, Islamabad, Pakistan (Remote)**
- 2024 **Tools:** Scratch, Python, App Development, Robotics
- Taught live classes in Scratch, Python, app development, and robotics to students across **Canada, the USA, and the UAE**.
 - Coached students preparing for **USACO** and **Bebras** computing competitions in the USA, UK, and Europe.
 - Mentored AI projects and designed project based lessons in computational thinking.

- Mar 2023 **Machine Learning Intern**, *NESCOM*, Islamabad, Pakistan
- Oct 2023
- Developed a deep learning ad detection and classification system, covering data collection, preprocessing, feature engineering, and evaluation.
 - Worked on signal processing and pattern recognition with a focus on accuracy.
- May 2022 **Deep Learning Intern**, *Bytewise Limited*, Islamabad, Pakistan
- Sep 2022
- Designed and trained computer vision and NLP models in TensorFlow and PyTorch.
 - Built custom datasets, augmentation pipelines, and classification and detection workflows.

Selected Projects

- 2026 **Juricratic: AI Litigation Intelligence**, *Personal Venture*, with [Connor](#), [juricratic.com](#)
- Present
- Litigation intelligence that models cases as **imperfect information games**, simulating outcomes with role bound **LLM agents** (judge, jury, opposing counsel, client).
- Nov 2024 **Multimodal Model Inference & Optimization Platform**, *Runware*, [runware.ai](#)
- Jul 2026
- Unified execution layer for image, video, audio, 3D, and LLM/VLM models with routing, batching, and VRAM aware scheduling.
- May 2026 **Multi Architecture LoRA Training Platform**, *Runware*, [runware.ai/lora-training](#)
- Jul 2026
- LoRA and QLoRA training for diffusion, LLM, and VLM models, from captioning and validation to **4 and 8 bit** memory efficient training and production hot swapping.
- Apr 2026 **vLLM Microservice Platform in Go**, *Runware*
- May 2026
- Go APIs over vLLM for prompt enhancement, captioning, age detection, and safety scoring, with Docker and Prometheus based **LLMOps**.
- 2026 **Low Latency Content Safety Model**, *Runware*
- Fine tuned a compact **vision transformer** from Freepik's NSFW detector for per category safety scores, running on every request with **subsecond** inference.
- Mar 2026 **Padelytix: Real Time Multi Camera Sports Perception**
- Aug 2026
- Player and ball tracking, clothing based reidentification, court homography, and highlight extraction, robust to occlusion and camera timing mismatch.
- Jul 2025 **Automated Lumber Takeoff and Cost Estimation from Construction Drawings**
- Feb 2026
- YOLO, SAM, and Detectron2 with Gemma and Qwen VLMs extract lumber types and quantities from drawings, served via FastAPI for automated bills of materials.
- Apr 2025 **Unilever × PSL 2025: Broadcast Logo Detection and Sponsorship Analytics**
- Jun 2025
- Real time YOLOv8 pipeline that detects, tracks, and grades sponsor logo visibility in live cricket, producing automated exposure and ROI reports.

Honors and Awards

- 2024 **Silver Medal**, *UET Taxila*, second highest aggregate in the session across all four years
- 2022 to 2023 **Laptop Award**, *Prime Minister's Youth Laptop Scheme*, Pakistan
- 2019 **1st Position in Pre Engineering**, *Year 12 (HSSC)*, college wide
- 2018 to 2020 **Full Tuition Fee Waiver and Stipend**, *Years 11 and 12 (HSSC)*, on Matric results

Technical Community & Leadership

- 2020 **Active Member**, *Google Developer Student Club (GDSC)*, UET Taxila
- 2024
- Organised workshops and led hands on sessions on TensorFlow, Firebase, and Flutter.
- 2020 **Active Member**, *Microsoft Learn Student Ambassadors (MLSA)*
- 2024
- Co organised talks and study jams on Azure and ML, and mentored peers on deploying ML projects.
- 2020 **Event Host**, *IEEE, UET Taxila Chapter*
- 2024
- Hosted seminars and panels connecting students with industry professionals in AI and software.
- 2020 **General Secretary**, *Umeed-e-Subh*, UET Taxila
- 2024
- Student run welfare community that established a school for underprivileged children across from UET Taxila.
 - Helped run daily classes and an evening IT drive teaching graphic design, basic coding, and video editing so children could earn while still in school.

Relevant Coursework

Mathematics	Calculus, Linear Algebra and Differential Equations, Probability and Statistics
ML & Vision	Machine Learning, Neural Networks, Computer Vision, Digital Image Processing
Systems	Data Structures and Algorithms, Operating Systems, Databases, Cloud Computing

Technical Skills

Programming	Python, C++, CUDA, Go, SQL, Unix Shell
ML	PyTorch, TensorFlow, Diffusers, Transformers, PEFT, TorchAO, LoRA and QLoRA, Diffusion and Flow Models, LLMs, VLMs, Knowledge Distillation
Train & Infer	FSDP, DTensor, Tensor and Context Parallelism, Quantization (FP8, FP4, INT8, INT4), Mixed Precision, FlashAttention, SageAttention, Ring and Sparse Attention, KV Caching, Diffusion Caching, PyTorch Compilation
GPU Systems	NVIDIA RTX, Hopper, Blackwell; AMD ROCm; Triton, CUTLASS, cuDNN, NCCL
Infrastructure	Distributed and Serverless Inference, Go Microservices, vLLM, LLMops, FastAPI, Docker, Kubernetes, Prometheus, AWS, RunPod, Vercel
Vision	YOLO, SAM, Detectron2, Tracking, Reidentification, Pose Estimation
Evaluation	Nsight Systems and Compute, Latency and Throughput Benchmarking, FID, CLIP Score, SSIM, PSNR, LPIPS

References

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